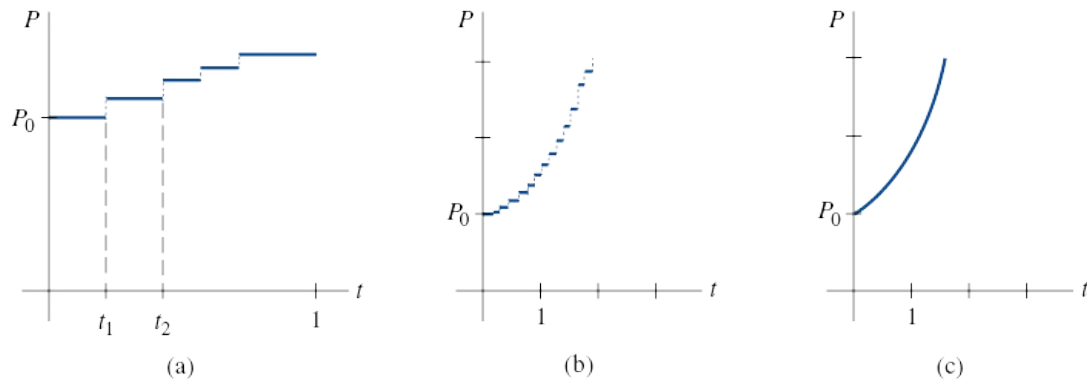


matter of convenience than of accuracy. However, for some purposes we may be satisfied if our model describes the system fairly closely when viewed macroscopically in time, as in Figures 2.7.7(b) and 2.7.7(c), rather than microscopically, as in Figure 2.7.7(a). Keep firmly in mind, a mathematical model is not reality.



**FIGURE 2.7.7** Population growth is a discrete process

## 2.7 Exercises

Answers to selected odd-numbered problems begin on page ANS-3.

## Growth and Decay

1. The population of a community is known to increase at a rate proportional to the number of people present at time  $t$ . If an initial population  $P_0$  has doubled in 5 years, how long will it take to triple? To quadruple?
2. Suppose it is known that the population of the community in Problem 1 is 10,000 after 3 years. What was the initial population  $P_0$ ? What will the population be in 10 years? How fast is the population growing at  $t = 10$ ?
3. The population of a town grows at a rate proportional to the population present at time  $t$ . The initial population of 500 increases by 15% in 10 years. What will the population be in 30 years? How fast is the population

growing at  $t = 30$ ?

4. The population of bacteria in a culture grows at a rate proportional to the number of bacteria present at time  $t$ . After 3 hours it is observed that 400 bacteria are present. After 10 hours 2000 bacteria are present. What was the initial number of bacteria?
5. The radioactive isotope of lead, Pb-209, decays at a rate proportional to the amount present at time  $t$  and has a half-life of 3.3 hours. If 1 gram of this isotope is present initially, how long will it take for 90% of the lead to decay?
6. Initially, 100 milligrams of a radioactive substance was present. After 6 hours the mass had decreased by 3%. If the rate of decay is proportional to the amount of the substance present at time  $t$ , find the amount remaining after 24 hours.
7. Determine the half-life of the radioactive substance described in Problem 6.
8. **(a)** Consider the initial-value problem  $dA/dt = kA$ ,  $A(0) = A_0$ , as the model for the decay of a radioactive substance. Show that, in general, the half-life  $T$  of the substance is  $T = -(\ln 2)/k$ .  
**(b)** Show that the solution of the initial-value problem in part (a) can be written  $A(t) = A_0 2^{-t/T}$ .  
**(c)** If a radioactive substance has a half-life  $T$  given in part (a), how long will it take an initial amount  $A_0$  of the substance to decay to  $\frac{1}{8}A_0$ ?
9. The amount  $A(t)$  of a radioactive substance remaining at any time can be determined directly from the initial amount  $A_0$  and the amount  $A_1$  of the substance present at a time  $t_1 > 0$ . Show that

$$A(t) = A_0 \left( \frac{A_1}{A_0} \right)^{t/t_1}.$$

- 10.** Using the result in Problem 9, show that the half-life  $T$  of a radioactive substance is

$$T = \frac{t_1 \ln 2}{\ln A_0 - \ln A_1}.$$

Use this result to verify the half-life in Problem 7.

- 11.** When a vertical beam of light passes through a transparent medium, the rate at which its intensity  $I$  decreases is proportional to  $I(t)$ , where  $t$  represents the thickness of the medium (in feet). In clear seawater, the intensity 3 feet below the surface is 25% of the initial intensity  $I_0$  of the incident beam. What is the intensity of the beam 15 feet below the surface?
- 12.** When interest is compounded continuously, the amount of money increases at a rate proportional to the amount  $S$  present at time  $t$ , that is,  $dS/dt = rS$ , where  $r$  is the annual rate of interest.
- (a)** Find the amount of money accrued at the end of 5 years when \$5000 is deposited in a savings account drawing  $5\frac{3}{4}\%$  annual interest compounded continuously.
- (b)** In how many years will the initial sum deposited have doubled?
- (c)** Use a calculator to compare the amount obtained in part (a) with the amount  $S = 5000(1 + \frac{1}{4}(0.0575))^{5(4)}$  that is accrued when interest is compounded quarterly.

## Carbon Dating

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- 13.** Archaeologists used pieces of burned wood, or charcoal, found at the site to date prehistoric paintings and drawings

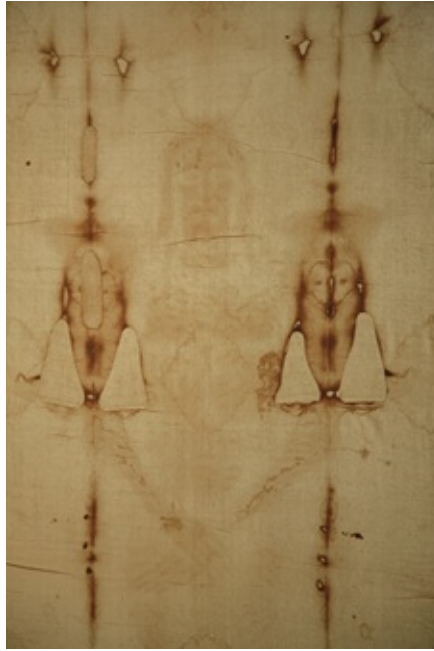
on walls and ceilings in a cave in Lascaux, France. See the photo below. Use the information on pages 78 and 79 to determine the approximate age of a piece of burned wood, if it was found that 85.5% of the C-14 found in living trees of the same type had decayed.



Cave wall painting in Problem 13

© thipjang/Shutterstock

- 14.** The **Shroud of Turin**, which shows the negative image of the body of a man who appears to have been crucified, is believed by many to be the burial shroud of Jesus of Nazareth. See the photo on the next page. In 1988 the Vatican granted permission to have the shroud carbon dated. Three independent scientific laboratories analyzed the cloth and concluded that the shroud was approximately 660 years old,<sup>\*</sup> an age consistent with its historical appearance. Using this age, determine what percentage of the original amount of C-14 remained in the cloth as of 1988.



Shroud image in Problem 14

© imageBROKER/Shutterstock

## Newton's Law of Cooling/Warming

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- 15.** A thermometer is removed from a room where the temperature is  $70^{\circ}\text{F}$  and is taken outside, where the air temperature is  $10^{\circ}\text{F}$ . After one-half minute the thermometer reads  $50^{\circ}\text{F}$ . What is the reading of the thermometer at  $t = 1$  min? How long will it take for the thermometer to reach  $15^{\circ}\text{F}$ ?
- 16.** A thermometer is taken from an inside room to the outside, where the air temperature is  $5^{\circ}\text{F}$ . After 1 minute the thermometer reads  $55^{\circ}\text{F}$ , and after 5 minutes it reads  $30^{\circ}\text{F}$ . What is the initial temperature of the inside room?
- 17.** A small metal bar, whose initial temperature was  $20^{\circ}\text{C}$ , is dropped into a large container of boiling water. How long will it take the bar to reach  $90^{\circ}\text{C}$  if it is known that its temperature increased  $2^{\circ}$  in 1 second? How long will it take the bar to reach  $98^{\circ}\text{C}$ ?

- 18.** Two large containers  $A$  and  $B$  of the same size are filled with different fluids. The fluids in containers  $A$  and  $B$  are maintained at  $0^{\circ}\text{C}$  and  $100^{\circ}\text{C}$ , respectively. A small metal bar, whose initial temperature is  $100^{\circ}\text{C}$ , is lowered into container  $A$ . After 1 minute the temperature of the bar is  $90^{\circ}\text{C}$ . After 2 minutes the bar is removed and instantly transferred to the other container. After 1 minute in container  $B$  the temperature of the bar rises  $10^{\circ}$ . How long, measured from the start of the entire process, will it take the bar to reach  $99.9^{\circ}\text{C}$ ?
- 19.** A thermometer reading  $70^{\circ}\text{F}$  is placed in an oven preheated to a constant temperature. Through a glass window in the oven door, an observer records that the thermometer read  $110^{\circ}\text{F}$  after  $\frac{1}{2}$  minute and  $145^{\circ}\text{F}$  after 1 minute. How hot is the oven?
- 20.** At  $t = 0$  a sealed test tube containing a chemical is immersed in a liquid bath. The initial temperature of the chemical in the test tube is  $80^{\circ}\text{F}$ . The liquid bath has a controlled temperature (measured in degrees Fahrenheit) given by  $T_m(t) = 100 - 40e^{-0.1t}$ ,  $t \geq 0$ , where  $t$  is measured in minutes.
- (a)** Assume that  $k = -0.1$  in (2). Before solving the IVP, describe in words what you expect the temperature  $T(t)$  of the chemical to be like in the short term. In the long term.
- (b)** Solve the initial-value problem. Use a graphing utility to plot the graph of  $T(t)$  on time intervals of various lengths. Do the graphs agree with your predictions in part (a)?
- 21.** A dead body was found within a closed room of a house where the temperature was a constant  $70^{\circ}\text{F}$ . At the time of discovery, the core temperature of the body was determined to be  $85^{\circ}\text{F}$ . One hour later a second measurement showed that the core temperature of the

body was 80°F. Assume that the time of death corresponds to  $t = 0$  and that the core temperature at that time was 98.6°F. Determine how many hours elapsed before the body was found.

22. Repeat Problem 21 if evidence indicated that the dead person was running a fever of 102°F at the time of death.

## Mixtures

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23. A tank contains 200 liters of fluid in which 30 grams of salt is dissolved. Brine containing 1 gram of salt per liter is then pumped into the tank at a rate of 4 L/min; the well-mixed solution is pumped out at the same rate. Find the number  $A(t)$  of grams of salt in the tank at time  $t$ .
24. Solve Problem 23 assuming that pure water is pumped into the tank.
25. A large tank is filled to capacity with 500 gallons of pure water. Brine containing 2 pounds of salt per gallon is pumped into the tank at a rate of 5 gal/min. The well-mixed solution is pumped out at the same rate. Find the number  $A(t)$  of pounds of salt in the tank at time  $t$ .
26. In Problem 25, what is the concentration  $c(t)$  of the salt in the tank at time  $t$ ? At  $t = 5$  min? What is the concentration of the salt in the tank after a long time; that is, as  $t \rightarrow \infty$ ? At what time is the concentration of the salt in the tank equal to one-half this limiting value?
27. Solve Problem 25 under the assumption that the solution is pumped out at a faster rate of 10 gal/min. When is the tank empty?
28. Determine the amount of salt in the tank at time  $t$  in Example 5 if the concentration of salt in the inflow is variable and given by  $c_{in}(t) = 2 + \sin(t/4)$  lb/gal. Without

actually graphing, conjecture what the solution curve of the IVP should look like. Then use a graphing utility to plot the graph of the solution on the interval  $[0, 300]$ . Repeat for the interval  $[0, 600]$  and compare your graph with that in Figure 2.7.4(a).

29. A large tank is partially filled with 100 gallons of fluid in which 10 pounds of salt is dissolved. Brine containing  $\frac{1}{2}$  pound of salt per gallon is pumped into the tank at a rate of 6 gal/min. The well-mixed solution is then pumped out at a slower rate of 4 gal/min. Find the number of pounds of salt in the tank after 30 minutes.
30. In Example 5 the size of the tank containing the salt mixture was not given. Suppose, as in the discussion following Example 5, that the rate at which brine is pumped into the tank is 3 gal/min but that the well-stirred solution is pumped out at a rate of 2 gal/min. It stands to reason that since brine is accumulating in the tank at the rate of 1 gal/min, any finite tank must eventually overflow. Now suppose that the tank has an open top and has a total capacity of 400 gallons.
  - (a) When will the tank overflow?
  - (b) What will be the number of pounds of salt in the tank at the instant it overflows?
  - (c) Assume that, although the tank is overflowing, the brine solution continues to be pumped in at a rate of 3 gal/min and the well-stirred solution continues to be pumped out at a rate of 2 gal/min. Devise a method for determining the number of pounds of salt in the tank at  $t = 150$  min.
  - (d) Determine the number of pounds of salt in the tank as  $t \rightarrow \infty$ . Does your answer agree with your intuition?
  - (e) Use a graphing utility to plot the graph  $A(t)$  on the interval  $[0, 500]$ .

## Series Circuits

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- 31.** A 30-volt electromotive force is applied to an  $LR$ -series circuit in which the inductance is 0.1 henry and the resistance is 50 ohms. Find the current  $i(t)$  if  $i(0) = 0$ . Determine the current as  $t \rightarrow \infty$ .
- 32.** Solve equation (8) under the assumption that  $E(t) = E_0 \sin \omega t$  and  $i(0) = i_0$ .
- 33.** A 100-volt electromotive force is applied to an  $RC$ -series circuit in which the resistance is 200 ohms and the capacitance is  $10^{-4}$  farad. Find the charge  $q(t)$  on the capacitor if  $q(0) = 0$ . Find the current  $i(t)$ .
- 34.** A 200-volt electromotive force is applied to an  $RC$ -series circuit in which the resistance is 1000 ohms and the capacitance is  $5 \times 10^{-6}$  farad. Find the charge  $q(t)$  on the capacitor if  $i(0) = 0.4$ . Determine the charge and current at  $t = 0.005$  s. Determine the charge as  $t \rightarrow \infty$ .
- 35.** An electromotive force

$$E(t) = \begin{cases} 120, & 0 \leq t \leq 20 \\ 0, & t > 20 \end{cases}$$

is applied to an  $LR$ -series circuit in which the inductance is 20 henries and the resistance is 2 ohms. Find the current  $i(t)$  if  $i(0) = 0$ .

- 36.** An  $LR$ -series circuit has a variable inductor with the inductance defined by

$$L(t) = \begin{cases} 1 - \frac{1}{10}t, & 0 \leq t < 10 \\ 0, & t > 10. \end{cases}$$

Find the current  $i(t)$  if the resistance is 0.2 ohm, the impressed voltage is  $E(t) = 4$ , and  $i(0) = 0$ . Graph  $i(t)$ .

## Additional Linear Models

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- 37. Air Resistance** In (14) of Section 1.3 we saw that a differential equation describing the velocity  $v$  of a falling mass subject to air resistance proportional to the instantaneous velocity is

$$m \frac{dv}{dt} = mg - kv,$$

where  $k > 0$  is a constant of proportionality called the drag coefficient. The positive direction is downward.

**(a)** Solve the equation subject to the initial condition  $v(0) = v_0$ .

**(b)** Use the solution in part (a) to determine the limiting, or terminal, velocity of the mass. We saw how to determine the terminal velocity without solving the DE in Problem 40 in Exercises 2.1.

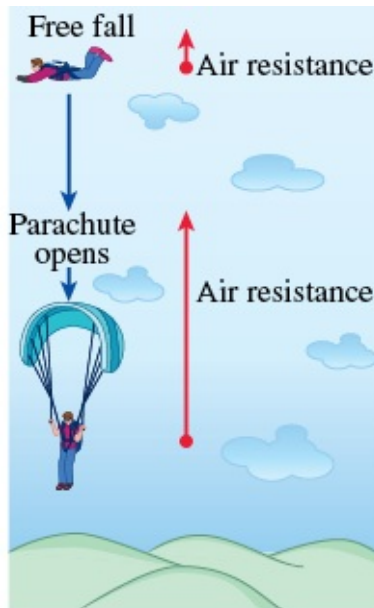
**(c)** If the distance  $s$ , measured from the point where the mass was released above ground, is related to velocity  $v$  by  $ds/dt = v$ , find an explicit expression for  $s(t)$  if  $s(0) = 0$ .

- 38. How High?—No Air Resistance** Suppose a small cannonball weighing 16 pounds is shot vertically upward with an initial velocity  $v_0 = 300$  ft/s. The answer to the question, “How high does the cannonball go?” depends on whether we take air resistance into account.

**(a)** Suppose air resistance is ignored. If the positive direction is upward, then a model for the state of the cannonball is given by  $d^2s/dt^2 = -g$  (equation (12) of Section 1.3). Since  $ds/dt = v(t)$  the last differential equation is the same as  $dv/dt = -g$ , where we take  $g = 32$  ft/s<sup>2</sup>. Find the velocity  $v(t)$  of the cannonball at time  $t$ .

(b) Use the result obtained in part (a) to determine the height  $s(t)$  of the cannonball measured from ground level. Find the maximum height attained by the cannonball.

- 39. How High?—Linear Air Resistance** Repeat Problem 38, but this time assume that air resistance is proportional to instantaneous velocity. It stands to reason that the maximum height attained by the cannonball must be *less* than that in part (b) of Problem 38. Show this by supposing that the drag coefficient is  $k = 0.0025$ . [*Hint*: Slightly modify the DE in Problem 37.]
- 40. Skydiving** A skydiver weighs 125 pounds, and her parachute and equipment combined weigh another 35 pounds. After exiting from a plane at an altitude of 15,000 feet, she waits 15 seconds and opens her parachute. See **FIGURE 2.7.8**. Assume the constant of proportionality in the model in Problem 37 has the value  $k = 0.5$  during free fall and  $k = 10$  after the parachute is opened. Assume that her initial velocity on leaving the plane is zero. What is her velocity and how far has she traveled 20 seconds after leaving the plane? How does her velocity at 20 seconds compare with her terminal velocity? How long does it take her to reach the ground? [*Hint*: Think in terms of two distinct IVPs.]



**FIGURE 2.7.8** Skydiver in Problem 40

- 41. Evaporating Raindrop** As a raindrop falls, it evaporates while retaining its spherical shape. If we make the further assumptions that the rate at which the raindrop evaporates is proportional to its surface area and that air resistance is negligible, then a model for the velocity  $v(t)$  of the raindrop is

$$\frac{dv}{dt} + \frac{3(k/\rho)}{(k/\rho)t + r_0} v = g.$$

Here  $\rho$  is the density of water,  $r_0$  is the radius of the raindrop at  $t = 0$ ,  $k < 0$  is the constant of proportionality, and the downward direction is taken to be the positive direction.

- (a)** Solve for  $v(t)$  if the raindrop falls from rest.
- (b)** Reread Problem 36 of Exercises 1.3 and then show that the radius of the raindrop at time  $t$  is  $r(t) = (k/\rho)t + r_0$ .
- (c)** If  $r_0 = 0.01$  ft and  $r = 0.007$  ft 10 seconds after the raindrop falls from a cloud, determine the time at which the raindrop has evaporated completely.

**42. Fluctuating Population** The differential equation  $dP/dt = (k \cos t)P$ , where  $k$  is a positive constant, is a mathematical model for a population  $P(t)$  that undergoes yearly seasonal fluctuations. Solve the equation subject to  $P(0) = P_0$ . Use a graphing utility to obtain the graph of the solution for different choices of  $P_0$ .

**43. Population Model** In one model of the changing population  $P(t)$  of a community, it is assumed that

$$\frac{dP}{dt} = \frac{dB}{dt} - \frac{dD}{dt},$$

where  $dB/dt$  and  $dD/dt$  are the birth and death rates, respectively.

(a) Solve for  $P(t)$  if  $dB/dt = k_1P$  and  $dD/dt = k_2P$ .

(b) Analyze the cases  $k_1 > k_2$ ,  $k_1 = k_2$ , and  $k_1 < k_2$ .

**44. Memorization** When forgetfulness is taken into account, the rate of memorization of a subject is given by

$$\frac{dA}{dt} = k_1(M - A) - k_2 A,$$

where  $k_1 > 0$ ,  $k_2 > 0$ ,  $A(t)$  is the amount to be memorized in time  $t$ ,  $M$  is the total amount to be memorized, and  $M - A$  is the amount remaining to be memorized. See Problems 25 and 26 in Exercises 1.3.

(a) Since the DE is autonomous, use the phase portrait concept of Section 2.1 to find the limiting value of  $A(t)$  as  $t \rightarrow \infty$ . Interpret the result.

(b) Solve for  $A(t)$  subject to  $A(0) = 0$ . Sketch the graph of  $A(t)$  and verify your prediction in part (a).

**45. Drug Dissemination** A mathematical model for the rate at which a drug disseminates into the bloodstream is given by  $dx/dt = r - kx$ , where  $r$  and  $k$  are positive constants. The function  $x(t)$  describes the concentration of the drug in

the bloodstream at time  $t$ .

(a) Since the DE is autonomous, use the phase portrait concept of Section 2.1 to find the limiting value of  $x(t)$  as  $t \rightarrow \infty$ .

(b) Solve the DE subject to  $x(0) = 0$ . Sketch the graph of  $x(t)$  and verify your prediction in part (a). At what time is the concentration one-half this limiting value?

- 46. Rocket Motion** Suppose a small single-stage rocket of total mass  $m(t)$  is launched vertically and that the rocket consumes its fuel at a constant rate. If the positive direction is upward and if we take air resistance to be linear, then a differential equation for its velocity  $v(t)$  is given by

$$\frac{dv}{dt} + \frac{k - \lambda}{m_0 - \lambda t} v = -g + \frac{R}{m_0 - \lambda t},$$

where  $k$  is the drag coefficient,  $\lambda$  is the rate at which fuel is consumed,  $R$  is the thrust of the rocket,  $m_0$  is the total mass of the rocket at  $t = 0$ , and  $g$  is the acceleration due to gravity. See Problem 21 in Exercises 1.3.

(a) Find the velocity  $v(t)$  of the rocket if  $m_0 = 200$  kg,  $R = 2000$  N,  $\lambda = 1$  kg/s,  $g = 9.8$  m/s<sup>2</sup>,  $k = 3$  kg/s, and  $v(0) = 0$ .

(b) Use  $ds/dt = v$  and the result in part (a) to find the height  $s(t)$  of the rocket at time  $t$ .

- 47. Rocket Motion—Continued** In Problem 46, suppose that of the rocket's initial mass  $m_0$ , 50 kg is the mass of the fuel.

(a) What is the burnout time  $t_b$ , or the time at which all the fuel is consumed? See Problem 22 in Exercises 1.3.

(b) What is the velocity of the rocket at burnout?

(c) What is the height of the rocket at burnout?

(d) Why would you expect the rocket to attain an altitude higher than the number in part (b)?

(e) After burnout what is a mathematical model for the velocity of the rocket?

## Discussion Problems

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- 48. Cooling and Warming** A small metal bar is removed from an oven whose temperature is a constant  $300^{\circ}\text{F}$  into a room whose temperature is a constant  $70^{\circ}\text{F}$ . Simultaneously, an identical metal bar is removed from the room and placed into the oven. Assume that time  $t$  is measured in minutes. Discuss: Why is there a future value of time, call it  $t^* > 0$ , at which the temperature of each bar is the same?
- 49. Heart Pacemaker** A heart pacemaker, shown in **FIGURE 2.7.9**, consists of a switch, a battery, a capacitor, and the heart as a resistor. When the switch  $S$  is at  $P$ , the capacitor charges; when  $S$  is at  $Q$ , the capacitor discharges, sending an electrical stimulus to the heart. In Problem 57 in Exercises 2.3, we saw that during the time the electrical stimulus is being applied to the heart, the voltage  $E$  across the heart satisfies the linear DE

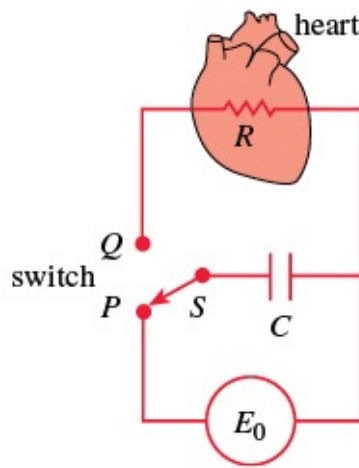
$$\frac{dE}{dt} = -\frac{1}{RC}E.$$

(a) Let us assume that over the time interval of length  $t_1$ ,  $(0, t_1)$ , the switch  $S$  is at position  $P$  shown in Figure 2.7.9 and the capacitor is being charged. When the switch is moved to position  $Q$  at time  $t_1$  the capacitor discharges, sending an impulse to the heart over the time interval of length  $t_2$ :  $[t_1, t_1 + t_2)$ . Thus, over the initial charging/discharging interval  $(0, t_1 + t_2)$  the voltage to the heart is actually modeled by the piecewise-linear differential equation

$$\frac{dE}{dt} = \begin{cases} 0, & 0 \leq t < t_1 \\ -\frac{1}{RC}E, & t_1 \leq t < t_1 + t_2. \end{cases}$$

By moving  $S$  between  $P$  and  $Q$ , the charging and discharging over time intervals of lengths  $t_1$  and  $t_2$  is repeated indefinitely. Suppose  $t_1 = 4$  s,  $t_2 = 2$  s,  $E_0 = 12$  V, and  $E(0) = 0$ ,  $E(4) = 12$ ,  $E(6) = 0$ ,  $E(10) = 12$ ,  $E(12) = 0$ , and so on. Solve for  $E(t)$  for  $0 \leq t \leq 24$ .

**(b)** Suppose for the sake of illustration that  $R = C = 1$ . Use a graphing utility to graph the solution for the IVP in part (a) for  $0 \leq t \leq 24$ .



**FIGURE 2.7.9** Model of a pacemaker in Problem 49

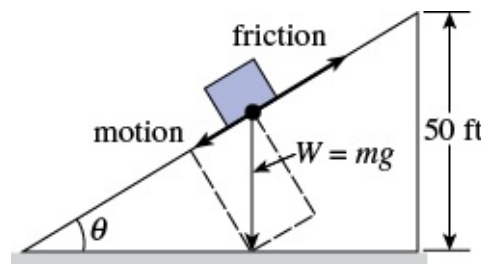
**50. Sliding Box** **(a)** A box of mass  $m$  slides down an inclined plane that makes an angle  $\theta$  with the horizontal as shown in **FIGURE 2.7.10**. Find a differential equation for the velocity  $v(t)$  of the box at time  $t$  in each of the following three cases:

- (i) No sliding friction and no air resistance
- (ii) With sliding friction and no air resistance
- (iii) With sliding friction and air resistance

In cases (ii) and (iii), use the fact that the force of friction opposing the motion of the box is  $\mu N$ , where  $\mu$  is the coefficient of sliding friction and  $N$  is the normal

component of the weight of the box. In case (iii) assume that air resistance is proportional to the instantaneous velocity.

**(b)** In part (a), suppose that the box weighs 96 lb, that the angle of inclination of the plane is  $\theta = 30^\circ$ , that the coefficient of sliding friction is  $\mu = \sqrt{3}/4$ , and that the additional retarding force due to air resistance is numerically equal to  $\frac{1}{4}v$ . Solve the differential equation in each of the three cases, assuming that the box starts from rest from the highest point 50 ft above ground.



**FIGURE 2.7.10** Box sliding down inclined plane in Problem 50

## Contributed Problem

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Canada

- 51. Air Exchange** A large room has a volume of  $2000 \text{ m}^3$ . The air in this room contains 0.25% by volume of carbon dioxide ( $\text{CO}_2$ ). Starting at 9:00 A.M. fresh air containing 0.04% by volume of  $\text{CO}_2$  is circulated into the room at the rate of  $400 \text{ m}^3/\text{min}$ . Assume that the stale air leaves the room at the same rate as the incoming fresh air and that the stale air and fresh air mix immediately in the room. See **FIGURE 2.7.11**.

- (a) If  $v(t)$  denotes the volume of  $\text{CO}_2$  in the room at time  $t$ , what is  $v(0)$ ? Find  $v(t)$  for  $t > 0$ . What is the percentage of  $\text{CO}_2$  in the air of the room at 9:05 A.M?
- (b) When does the air in the room contain 0.06% by volume of  $\text{CO}_2$ ?
- (c) What should be the flow rate of the incoming fresh air if it is required to reduce the level of  $\text{CO}_2$  in the room to 0.08% in 4 minutes?

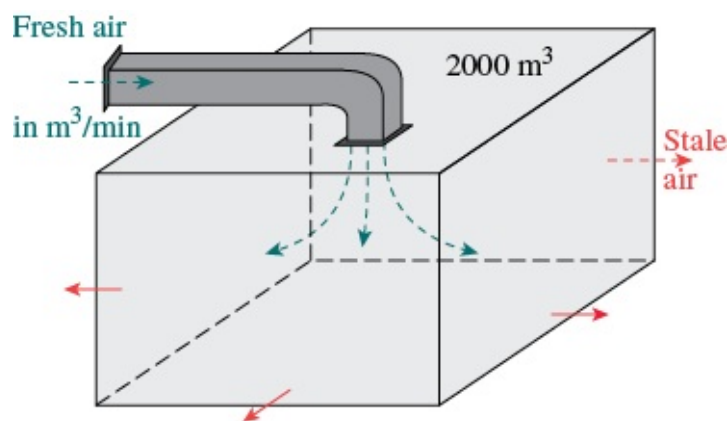


FIGURE 2.7.11 Air exchange in Problem 51

## Computer Lab Assignments

- 52. Sliding Box—Continued** (a) In Problem 50, let  $s(t)$  be the distance measured down the inclined plane from the highest point. Use  $ds/dt = v(t)$  and the solution for each of the three cases in part (b) of Problem 50 to find the time that it takes the box to slide completely down the inclined plane. A root-finding application of a CAS may be useful here.
- (b) In the case in which there is friction ( $\mu \neq 0$ ) but no air resistance, explain why the box will not slide down the plane starting from rest from the highest point above ground when the inclination angle  $\theta$  satisfies  $\tan \theta \leq \mu$ .

(c) The box *will* slide downward on the plane when  $\tan \theta \leq \mu$  if it is given an initial velocity  $v(0) = v_0 > 0$ . Suppose that  $\mu = \sqrt{3}/4$  and  $\theta = 23^\circ$ . Verify that  $\tan \theta \leq \mu$ . How far will the box slide down the plane if  $v_0 = 1$  ft/s?

(d) Using the values  $\mu = \sqrt{3}/4$  and  $\theta = 23^\circ$ , approximate the smallest initial velocity  $v_0$  that can be given to the box so that, starting at the highest point 50 ft above ground, it will slide completely down the inclined plane. Then find the corresponding time it takes to slide down the plane.

- 53. What Goes Up** (a) It is well known that the model in which air resistance is ignored, part (a) of Problem 38, predicts that the time  $t_a$  it takes the cannonball to attain its maximum height is the same as the time  $t_d$  it takes the cannonball to fall from the maximum height to the ground. Moreover, the magnitude of the impact velocity  $v_i$  will be the same as the initial velocity  $v_0$  of the cannonball. Verify both of these results.
- (b) Then, using the model in Problem 39 that takes linear air resistance into account, compare the value of  $t_a$  with  $t_d$  and the value of the magnitude of  $v_i$  with  $v_0$ . A root-finding application of a CAS (or graphic calculator) may be useful here.

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\*Some scholars have disagreed with the finding. For more information on this fascinating mystery, see the Shroud of Turin website home page at [www.shroud.com](http://www.shroud.com).